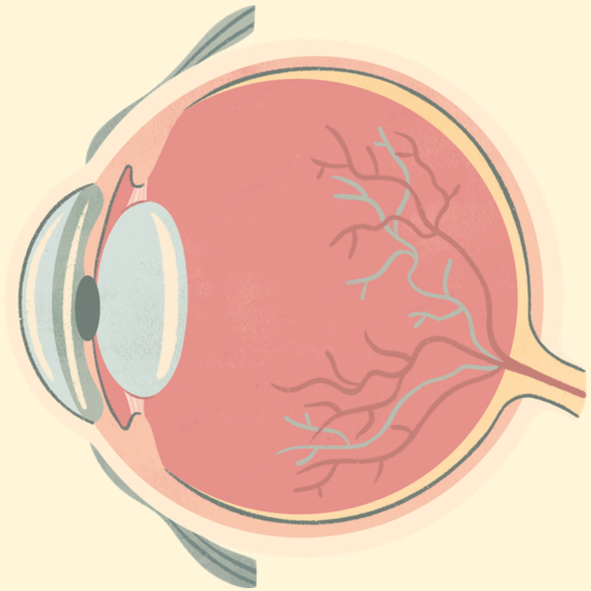
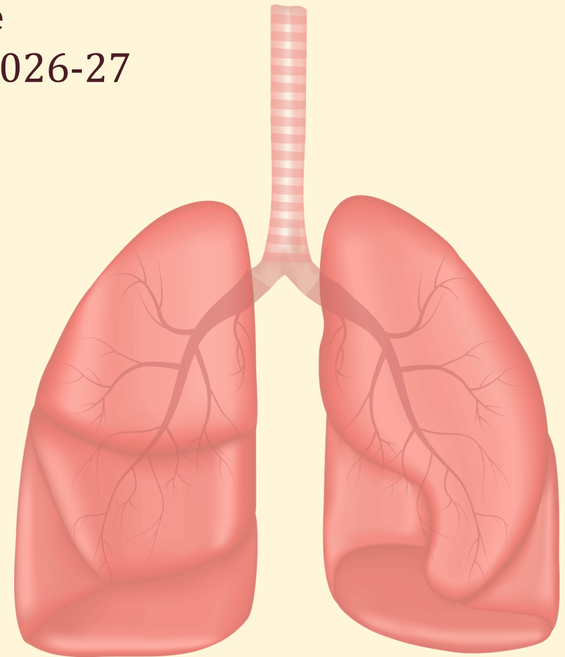


HYBRID DENSENET-121 AND QUANTUM CIRCUIT CLASSIFICATION OF MEDICAL IMAGES

Diabetic retinopathy, LIDC lung and Y-Net Capsule experiments

Course: Deep Learning | Guide: Dr. Ananya Bhattacharjee
Dept. of AI & ML, Symbiosis Institute of Technology, Pune | AY 2026-27

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Abstract and Objectives

DenseNet-121 classifier

4-qubit VQC on its features

Uncertainty layer: manual review

Grad-CAM++ heatmaps

80.91%

DenseNet-121 accuracy
DR (DDR), 5 grades, 3,725 test images, QWK 0.8129

85.42%

DenseNet-121 accuracy
LIDC lung, 3 classes, 48 test images

98.92%

DenseNet-121 accuracy
Y-Net Capsule, 3 classes, 93 test images

- The VQC (39 trainable parameters) matches or trails the DenseNet in every experiment; fusion added nothing measurable
- Checkpoints chosen by validation macro F1 only; leakage-aware splits so copies of one source image never cross splits
- One seed per experiment, small 3-class test sets, and a Capsule image-size difference between classes call for careful reading

Project Overview: Three Experiments

Experiment	Data	Classes	Input	Train / Val / Test	GPU
1. DR	DDR retinal images, CLAHE	5 ordered grades	512 x 512	6,210 / 2,487 / 3,725	Tesla T4
2. LIDC	1,924 files, 319 source images	benign, malignant, normal	224 x 224	223 / 48 / 48	Tesla T4
3. Capsule	1,608 files, 615 source images	benign, malignant, normal	224 x 224	430 / 92 / 93	A100 40 GB

Shared Methodology

DenseNet-121 embedding (1,024-d)

Standardize, PCA to 4 components

Rescale to 0 to pi

4 qubits: RY embedding + 2 StronglyEntanglingLayers

Pauli-Z, linear layer 4 to 3

VQC: 24 circuit weights + 15 linear = 39 trainable parameters; 4 components keep only 35.28% (LIDC) and 36.18% (Capsule) of embedding variance

Training

- AdamW, batch 16, 100 epochs, seed 42
- Class weights: inverse sqrt of counts
- Checkpoint by validation macro F1

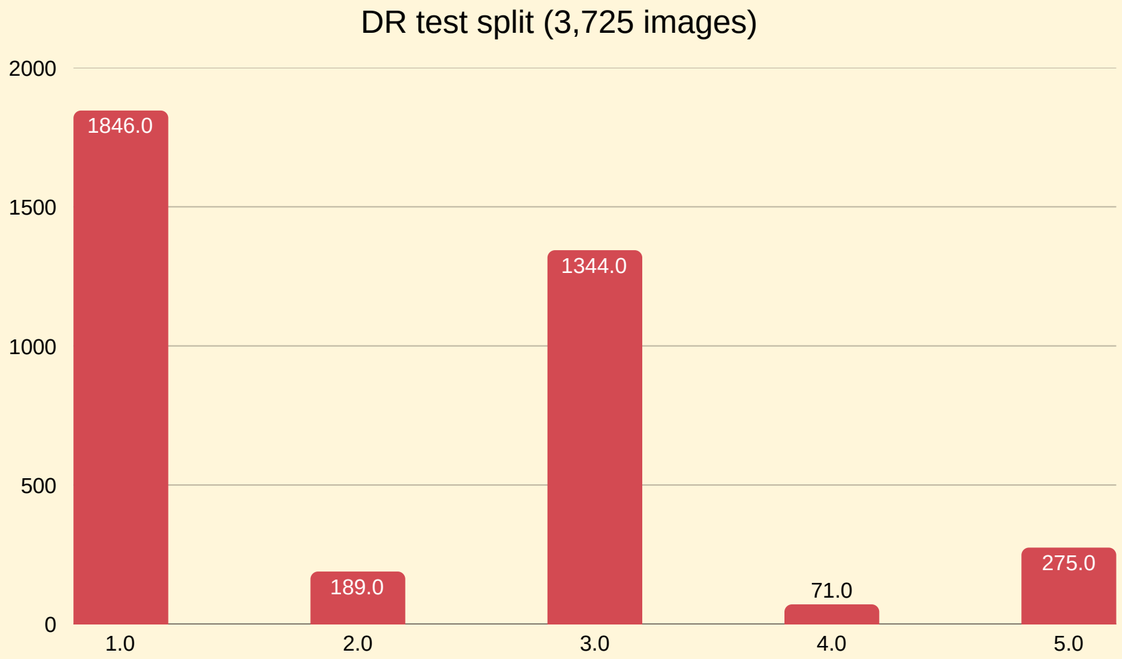
Fusion and uncertainty

- w x P(DenseNet) + (1-w) x P(VQC), w 0.10 to 0.90 on validation
- LOW_CONFIDENCE, HIGH_UNCERTAINTY (+ GRADE_INCONSISTENCY for DR); thresholds illustrative

Explain and measure

- Grad-CAM++ on features.denseblock4
- Accuracy, macro and weighted F1, per-class, confusion matrices
- QWK for DR only

Experiment 1: DR Grading (DDR) - Data and Training



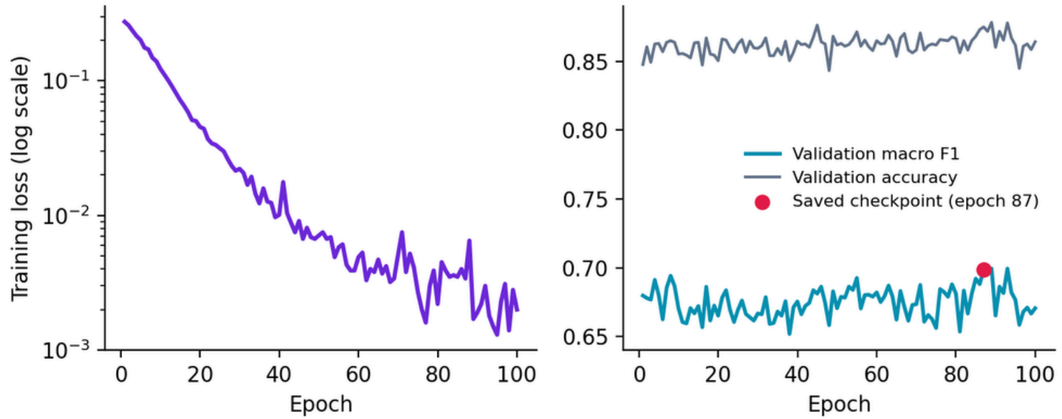
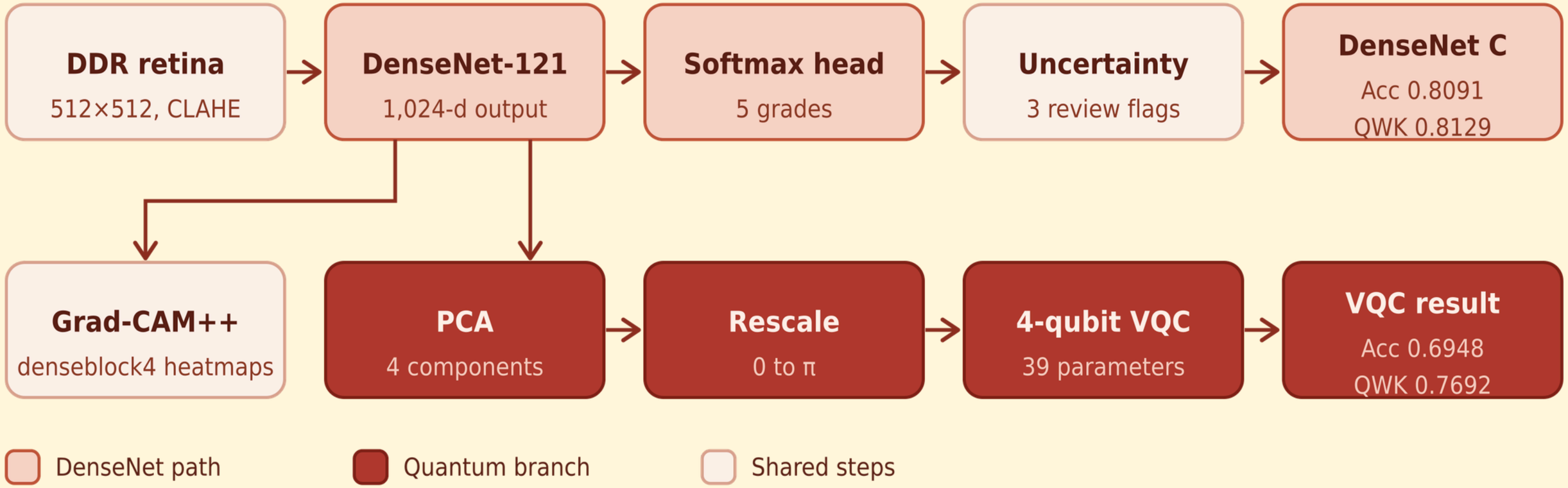


Figure 1. DenseNet-121 C fine-tuning history

- 12,422 images split about 50/20/30; cached 512 x 512 with CLAHE; Severe has only 71 test images, so one image moves its recall 1.4 points
- Fine-tuning DenseNet C: ~163 s/epoch, 100 epochs (~4.5 h); train loss 0.2747 to 0.0020 while validation macro F1 stayed 0.65 to 0.70
- Final checkpoint: epoch 87, validation macro F1 0.6985

Architectural Diagram

Experiment 1: DR grading (DDR, 5 grades)



DR Test Results (3,725 images)

Model	Accuracy	Macro precision	Macro recall	Macro F1	Weighted F1	QWK
DenseNet B	0.8013	0.7079	0.5872	0.6146	0.7808	0.8002
DenseNet C	0.8091	0.7058	0.6271	0.6579	0.8003	0.8129
VQC	0.6948	0.5649	0.6632	0.5711	0.7091	0.7692

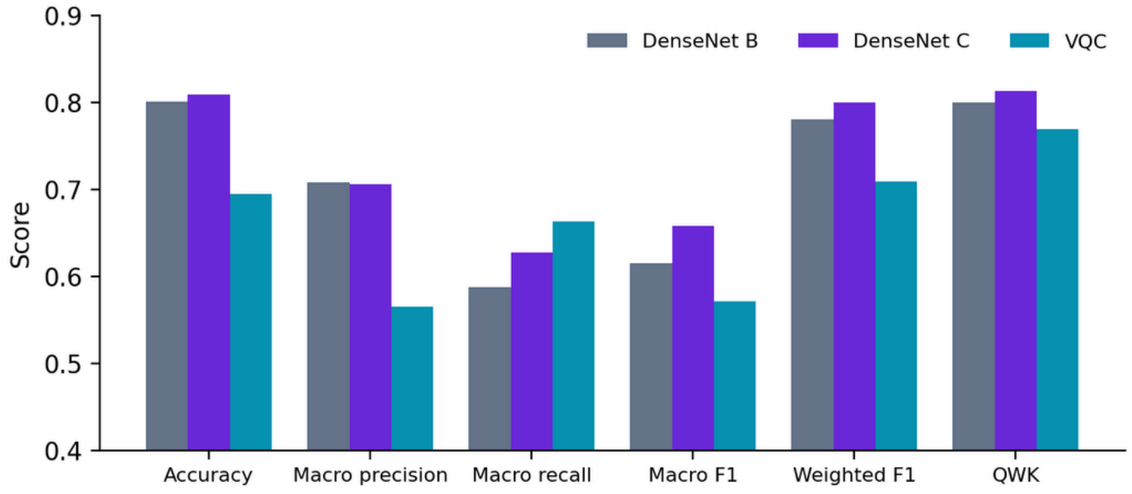


Figure 2. DR test metrics for the three models

- DenseNet C leads accuracy, macro F1, weighted F1 and QWK
- VQC has the highest macro recall (0.6632) but the lowest accuracy (0.6948): more rare-grade finds, many false alarms
- Fine-tuning vs earlier result: accuracy +0.0217, macro F1 +0.0119, QWK -0.0120

DR Per-Class Behaviour

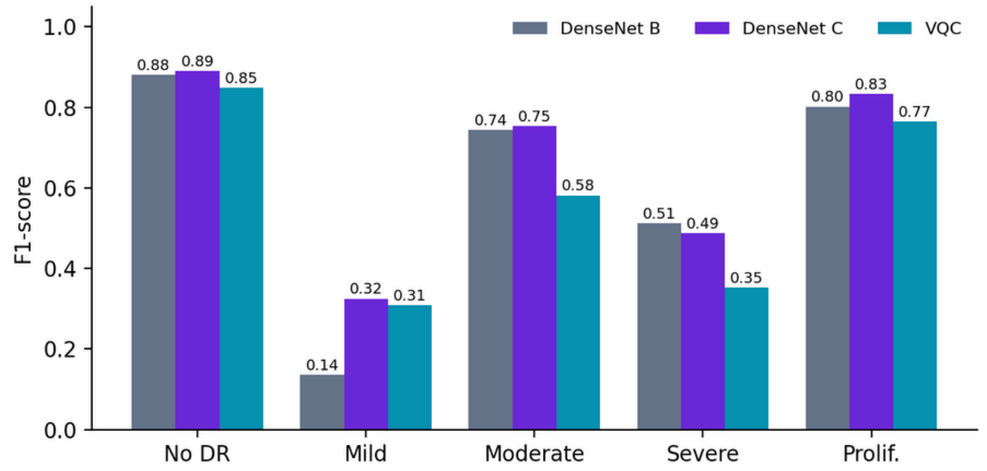
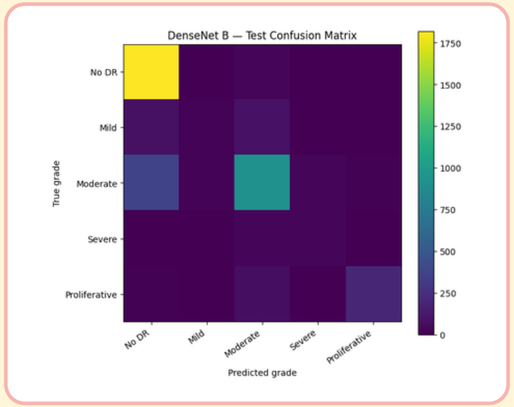
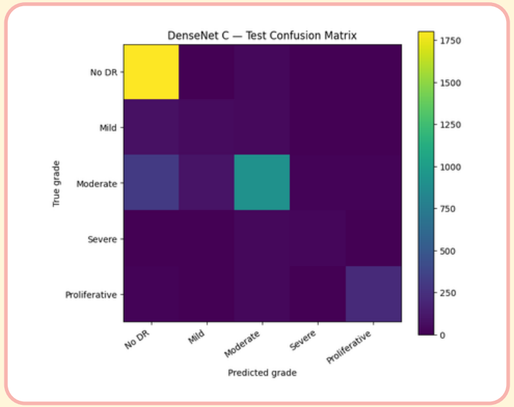
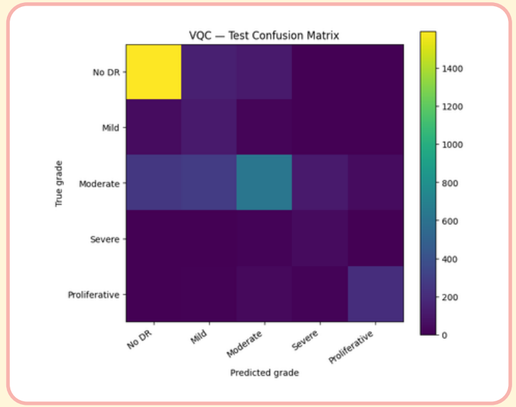


Figure 3. DR per-class F1 (test)

- No DR and Proliferative handled well (F1 0.80 to 0.89)
- Mild is weakest: DenseNet C lifts Mild F1 0.1351 to 0.3246 and recall 7.94% to 29.63% (56 of 189 vs 15)
- VQC finds the most Mild (114) and Severe (44) images, but precision is only 0.2073 and 0.2458







Largest DenseNet error block: Moderate predicted as No DR. The VQC also spreads Moderate across Mild, Severe and Proliferative (Figure 4).

Experiment 2: LIDC Lung Images (48 test images)

Model	Accuracy	Macro precision	Macro recall	Macro F1	Weighted F1
DenseNet-121	0.8542	0.8611	0.8601	0.8494	0.8476
VQC	0.8542	0.8543	0.8575	0.8521	0.8513
DenseNet + VQC (0.90 / 0.10)	0.8542	0.8611	0.8601	0.8494	0.8476

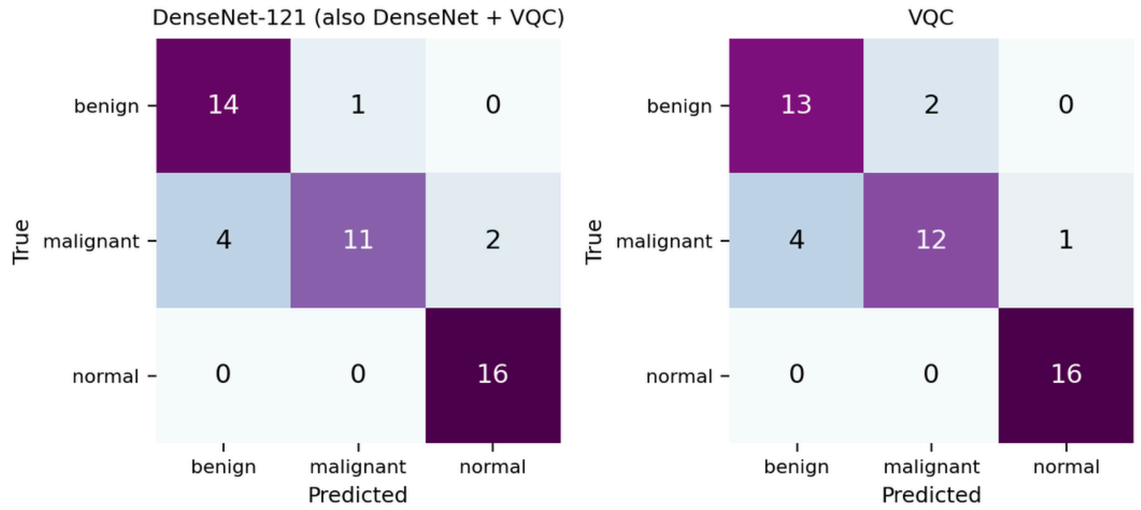
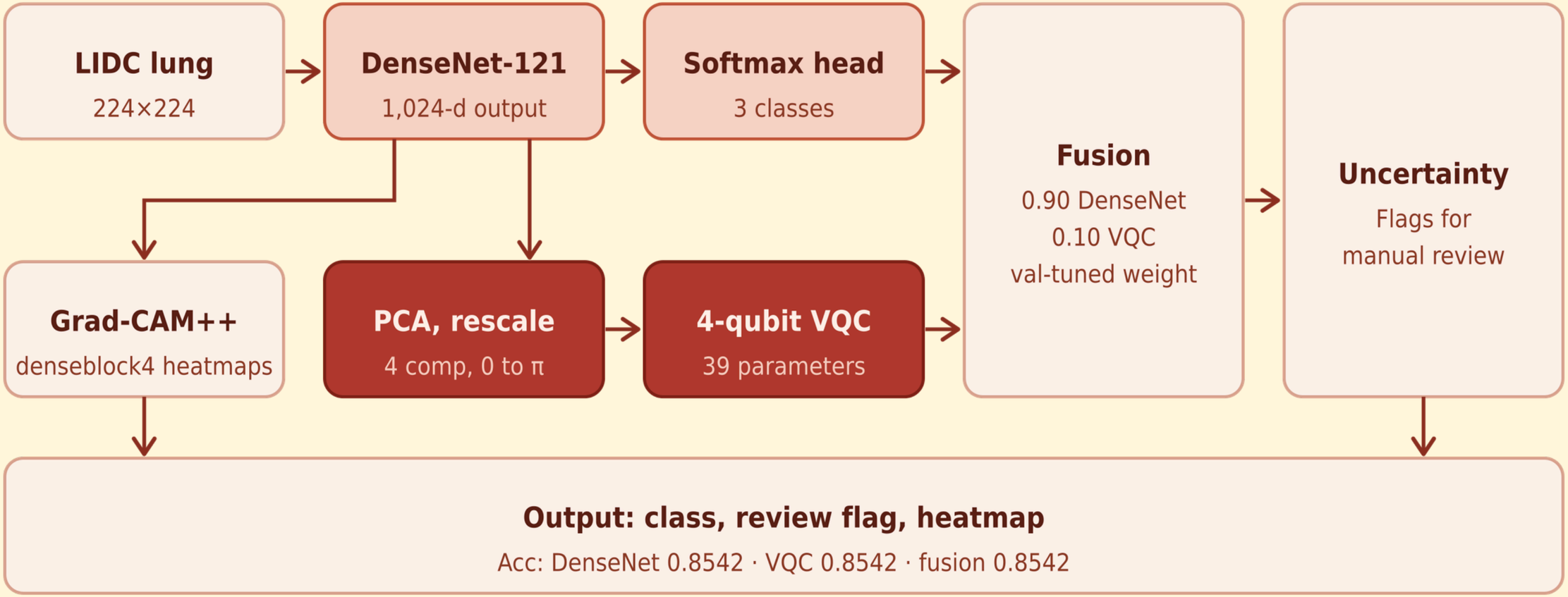


Figure 8. LIDC test confusion matrices

- 1,924 files reduced to 319 source images, split 70/15/15 by source ID (223/48/48); ID 11_29 excluded
- All models make 7 errors of 48; fusion changed no prediction
- Malignant weakest: recall 64.71% (DenseNet), 70.59% (VQC); no normal image missed
- One test image = 2.1 points

Architectural Diagram

Experiment 2: LIDC lung (3 classes)



Experiment 3: Y-Net Capsule (93 test images)

Model	Accuracy	Macro precision	Macro recall	Macro F1	Weighted F1
DenseNet-121	0.9892	0.9792	0.9778	0.9778	0.9892
VQC	0.9785	0.9740	0.9556	0.9628	0.9779
DenseNet + VQC (0.10 / 0.90)	0.9785	0.9740	0.9556	0.9628	0.9779

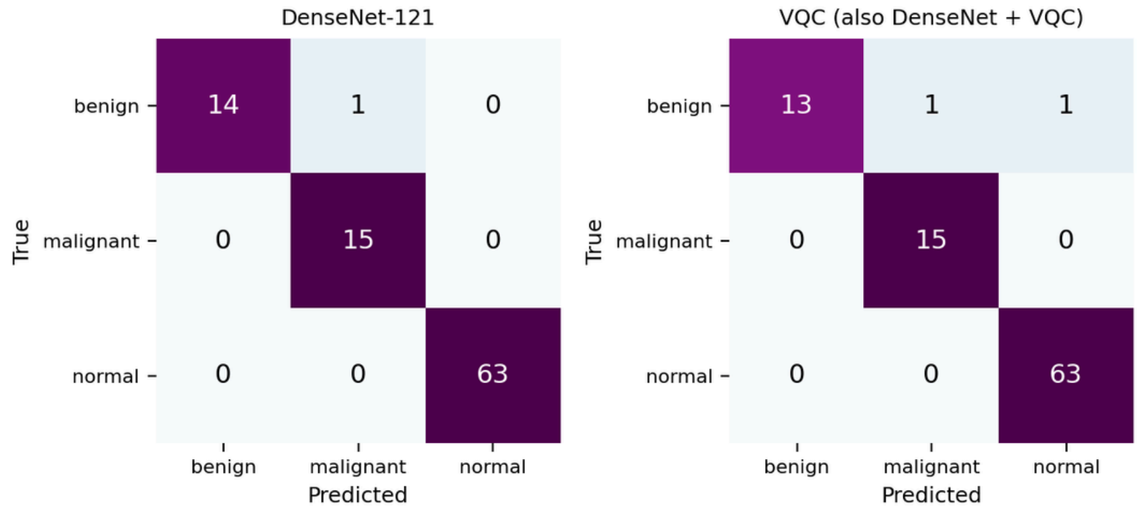
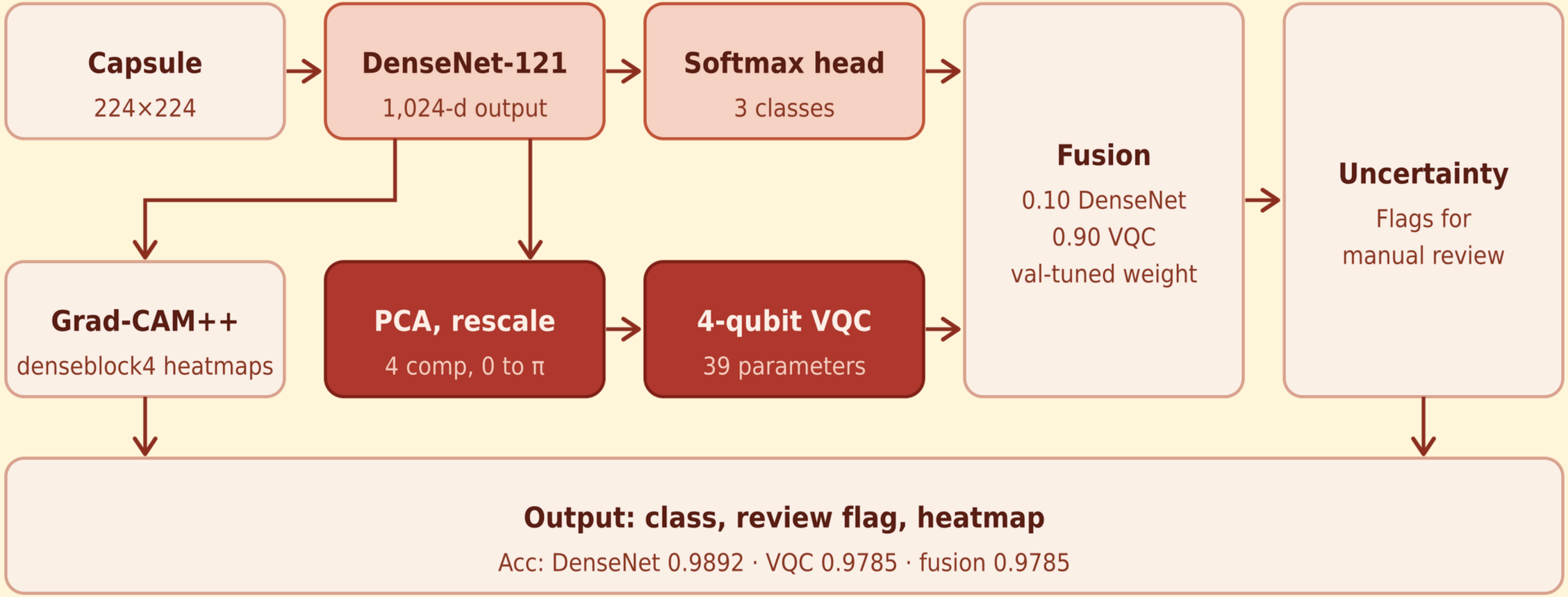


Figure 12. Capsule test confusion matrices

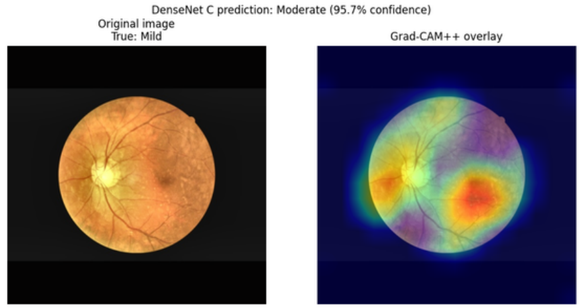
- 4 source IDs leaked across Train/Val, so the archive was pooled and re-split 430/92/93 (15 benign, 15 malignant, 63 normal in test)
- DenseNet makes 1 error; VQC and fusion 2; only benign images missed
- Normal images are 512 x 512, others 256 x 256: a possible shortcut, so read the scores with care

Architectural Diagram

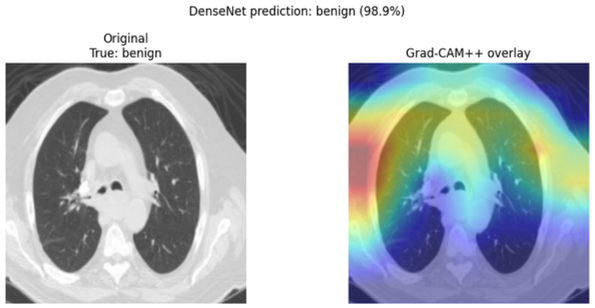
Experiment 3: Y-Net Capsule (3 classes)



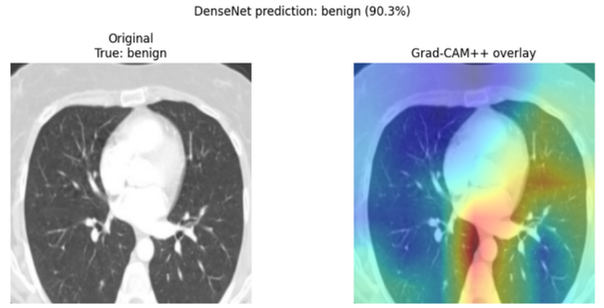
Uncertainty Layer and Grad-CAM++



DR: Mild predicted as Moderate (95.7%)



LIDC: benign, correct (98.9%)



Capsule: benign, correct (90.3%)

Experiment	Flagged for review	What the first ten rows show
DR	97 of 3,725 (2.60%)	2 wrong, neither flagged: a confident error passes the rules
LIDC	3 of 48 (6.25%)	All correct and unflagged
Capsule	6 of 93 (6.45%)	The one error (444-112, confidence 0.5190) is flagged

Heat on borders, the chest wall or scanner-table framing means the maps are not clinical evidence; no clinician reviewed them.

Cross-Experiment Comparison

Experiment	Test images	DenseNet-121 (acc / F1)	VQC (acc / F1)	Fusion
DR (DDR)	3,725	0.8091 / 0.6579 (model C)	0.6948 / 0.5711	Not used
LIDC lung	48	0.8542 / 0.8494	0.8542 / 0.8521	Same as DenseNet
Capsule	93	0.9892 / 0.9778	0.9785 / 0.9628	Same as VQC

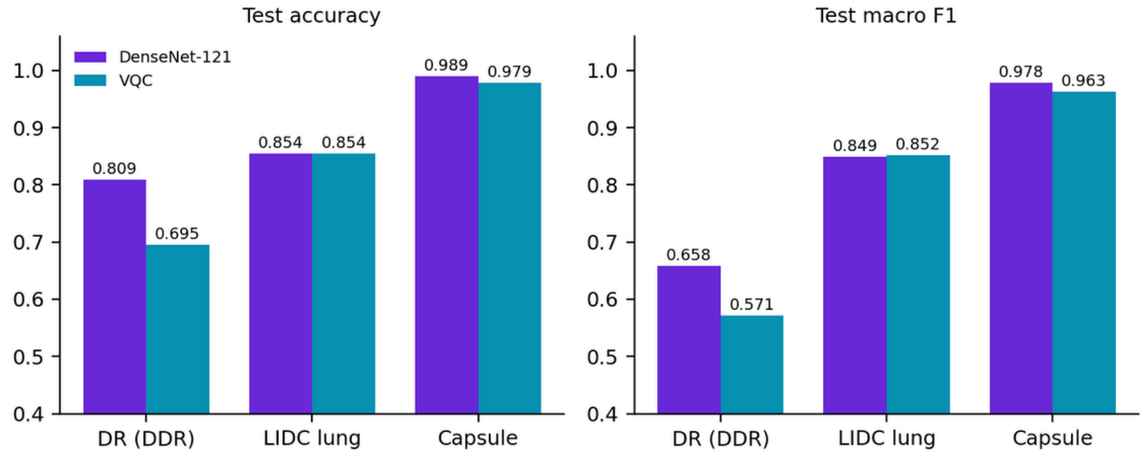


Figure 14. Test accuracy and macro F1

- VQC beats the DenseNet nowhere: 11.4 accuracy points behind on DR, tied on LIDC, one image behind on Capsule
- Fusion matched one of its inputs; validation sets were saturated
- Minority classes are the hard cases: DR Mild, LIDC malignant, Capsule benign
- Not a controlled comparison

Limitations and Next Steps

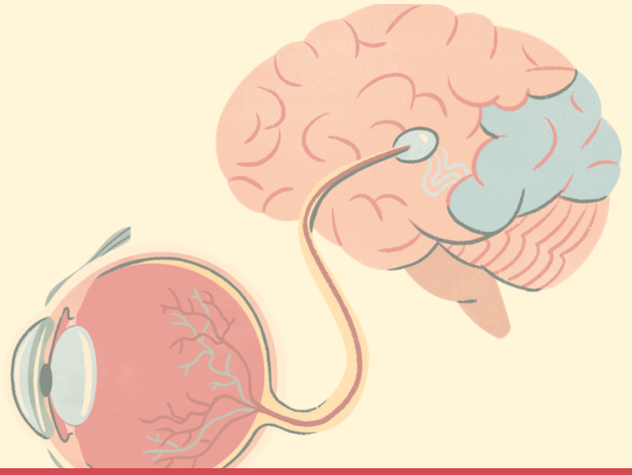
Limitations

- One seed and one test set per experiment; no confidence intervals
- Small LIDC/Capsule test sets; validation macro F1 saturates at 1.0
- No patient IDs: splits group by source ID only
- Capsule class size and framing differences; no segmentation masks
- Uncertainty thresholds untuned; confident errors pass

Next steps

- Patient-level splits
- Repeated seeds or cross-validation
- Larger test sets, masks

Research prototype: must not guide medical decisions. Colab notebooks, seed 42.



Thank You

Questions?

